

Kanika Chaudhary

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PROFESSIONAL SUMMARY

Computer Science and Engineering student and Gen AI Intern with a bias for action — independently designed, built, and shipped two live, publicly deployed products end-to-end, using AI tools and agent workflows to move fast without compromising quality. Comfortable picking up any tech stack needed to solve the problem at hand, from LLM agents to embedded IoT systems. Strong problem-solving and structured thinking, with a track record of owning ambiguous, real-world problems from prototype to working output.

EDUCATION

B.Tech, Computer Science & Engineering (AI & ML) — UPES, Dehradun	CGPA: 9.1 / 10
Class XII (CBSE) — ASASVM, Roorkee	79%
Class X (CBSE) — ASASVM, Roorkee	91%

TECHNICAL SKILLS

AI / GenAI (hands-on): LLM Agents, RAG Pipelines, Tool Calling, Prompt Engineering, LangChain, MCP, Ollama

Programming: Python, C++, Java

ML / Data: XGBoost, CNN, TensorFlow, scikit-learn, Pandas, NumPy

Backend & APIs: FastAPI, REST APIs, JWT Authentication

Databases: PostgreSQL, SQL, ChromaDB (Vector DB)

Frontend: React, Vite, Tailwind CSS

Tools & Platforms: Git, Docker, MQTT, Arduino/ESP32

INTERNSHIP EXPERIENCE

Intern — Naiyo24 Pvt. Ltd. May – July 2026

- Designed and orchestrated multi-step LLM agent workflows with tool-calling logic to automate complex, multi-stage tasks end-to-end, from prototype to working output.
- Applied backend API design and structured-data validation practices — the same principles used to build and secure production APIs independently in Secure Vault (FastAPI + PostgreSQL, JWT auth) — to improve the reliability of AI-driven workflow outputs.
- Built and evaluated XGBoost and CNN models alongside LLM-based components across structured and image data, improving output accuracy and reliability.

PROJECTS

MNEMOSYNE [↗](#) | [Python](#), [Git Hooks](#), [Cognee](#), [Knowledge Graph](#), [Ollama](#) | [CODE](#) | [LIVE](#) 2026

- Independently built and shipped an autonomous AI agent — a Git pre-commit hook that queries a memory layer before every commit — as a working, deployed end-to-end product, not a prototype.
- Designed a RAG-style retrieval system combining a Knowledge Graph (relationship storage) with a Vector Database (semantic similarity search) to surface past bug fixes, coding rules, and architectural decisions.
- Integrated a local LLM via Ollama (Llama 3.1) for reasoning over retrieved context, creating a persistent, queryable agent memory system that warns developers of recurring issues before they commit — solving a real workflow problem, not a toy demo.

SECURE VAULT [↗](#) | [React](#), [FastAPI](#), [PostgreSQL](#), [JWT](#), [AES](#) | [CODE](#) | [LIVE](#) 2025

- Independently designed and deployed a full-stack secure web application end-to-end — from architecture design through production deployment — enabling encrypted note and file storage with JWT-based auth and refresh tokens.
- Engineered AES encryption, bcrypt password hashing, and API protections (rate limiting, request validation, input sanitization, secure headers) to prevent SQL injection and XSS attacks.
- Designed a layered FastAPI + PostgreSQL backend (routers, services, middleware) and a React + Tailwind CSS dashboard with activity logging — demonstrating independent, fast, end-to-end product delivery.

SMART SOIL & FERTILIZER RECOMMENDATION [↗](#) | [Arduino](#), [ESP32](#), [MQTT](#) ([HiveMQ](#)), [XGBoost](#) | [DEMO](#) 2025

- Built an IoT-based system using Arduino and ESP32 to collect real-time soil parameters — NPK, temperature, and humidity — from field sensors, picking up embedded hardware as a new stack to solve the problem at hand.
- Implemented MQTT-based communication to stream live sensor data to the HiveMQ cloud platform for centralized monitoring.
- Integrated a trained XGBoost model to analyze soil parameters and recommend optimal fertilizer composition, closing the loop from raw sensor data to actionable output.

CERTIFICATIONS & ACCOMPLISHMENTS

PwC Advisory Launchpad Learning Program Feb 2026 – Present

- Selected for PwC's structured advisory learning program focusing on consulting fundamentals, business problem-solving, and data-driven decision making.